MONASH University

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ASP3231 Assignment 2
Semester One 2026
Faculty of Science
School of Physics and Astronomy

<u>AUTHORISED MATERIALS</u>

OPEN BOOK	**YES** ☒	**NO** ☐
CALCULATORS	**YES** ☒	**NO** ☐
SPECIFICALLY PERMITTED ITEMS	**YES** ☒	**NO** ☐

Question 1 (20 marks)

The 10.5-metre Keck telescopes are equipped with laser guide star adaptive optics (LGSAO) systems that are used in conjunction with near-infrared imagers and spectrographs. Technical and user manuals for the Keck LGSAO systems are available online at https://www2.keck.hawaii.edu/optics/lgsao/.

(a) (8 marks) For bright tip-tilt guide stars, the Keck LGSAO system can achieve a full width half maximum (FWHM) of 55 milliarcseconds and a Strehl ratio of 0.4. For a wavelength of 2.15 microns, how do these two numbers compare to the ideal diffraction limited solution.

(b) (6 marks) The Keck LGSAO system can function with tip-tilt guide stars as faint as $m_R = 18$. Why is the Keck LGSAO system able to function with such faint guide stars while systems using natural guide stars are limited to very bright stars? Limit your answer to no more than four sentences.

(c) (4 marks) At Keck typical value of r_0 is 25 cm and the turbulence is typically at 1800 metres altitude. At a wavelength of 2.15 micron what is the approximate size of the isoplanatic patch?

(d) (2 marks) The Keck telescopes can achieve higher angular resolution than James Webb Space Telescope, but cannot successfully obtain spectra of redshift $z > 10$ galaxies. In no more than two sentences, provide one reason why this is the case.

Question 2 (16 marks)

EP240414a is an unusual transient detected with the *Einstein Probe* (EP) satellite mission. The 90% uncertainty region (with a radius of 10″) is shown as the green circle in Figure 1, overlaid on an optical image of the field. Within the uncertainty region are a candidate host galaxy, an optical transient (first detected ≈ 3.13 hr after EP240414a), and a foreground star (known to be much closer to us thanks to a *Gaia* parallax distance).

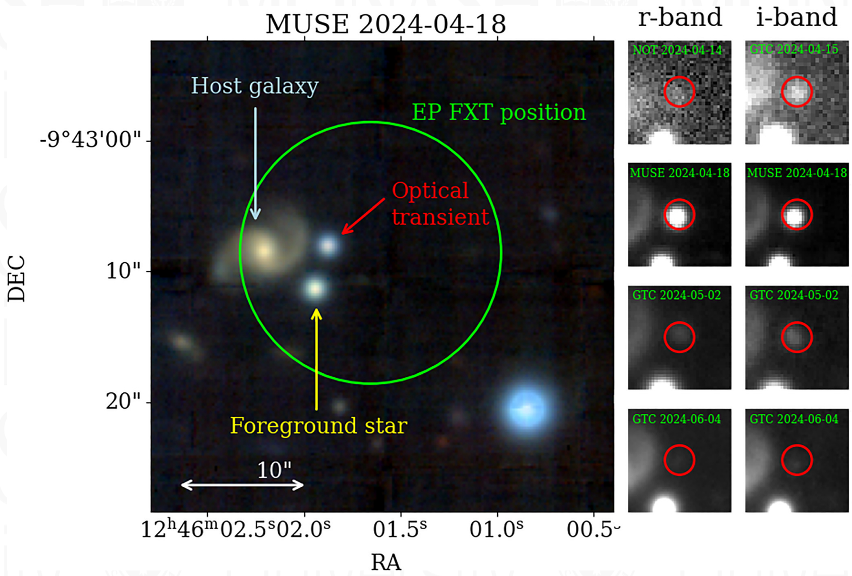

Figure 1: False-colour optical image from the MUSE observation, showing the *Einstein Probe* (EP) position for the transient, with 10″ radius at 90% confidence level (C.L.). The coincident optical transient, suggested host galaxy and an unrelated foreground star, all within the 90% C.L. region, are marked. The small panels show r- and i-band imaging at four different epochs. This figure is also available via https://tinyurl.com/9vvrn87n.

(a) (1 mark) The EP instruments are sensitive to photons with energies 0.5–4 keV. What part of the electromagnetic spectrum does this correspond to?

(b) (2 marks) What is the typical energy of a photon that the EP instruments might detect? Give your answer in erg.

(c) (4 marks) The EP wide-field telescope has a field-of-view of 3600 deg^2 and an effective area of approx. 1.5 cm^2 over this area. At peak the transient accumulates 124 counts (after background subtraction) in a 100 s exposure. Estimate the peak flux. Give your answer in erg cm^{-2} s^{-1}

(d) (4 marks) The typical background rate during wide-field telescope observations is 0.3 count s^{-1}. What is the 68% (1σ) uncertainty on the flux calculated in part (c)?

(e) (5 marks) Estimate the spectral flux density for EP240414a at its peak. Give your answer in Jy.

Question 3 (10 marks)

This question continues the discussion of EP240414a, as introduced in Question 2.

The claim is made that "since both the optical transient and the galaxy fall within the 90% confidence region, the probability of their association with EP240414a is also 90%".

(a) (3 marks) What is wrong with this claim?

List up to 3 reasons why this claim is incorrect, with no more than 2 sentences per reason.

(b) (4 marks) The average spatial density of galaxies in this region is 0.5 arcmin^{-2}. Estimate the probability the labeled galaxy is *not* associated with EP240414.

(c) (3 marks) Describe in your own words how the *time* coincidence of EP240414 and the optical transient might contribute to an association probability calculation, as in part (b).

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Useful information

Physical Constants:

$G =$	gravitational constant	$= 6.673 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}} = 6.673 \times 10^{-8}\ \mathrm{cm^3\,g^{-1}\,s^{-2}}$
$\Re =$	universal gas constant	$= 8.314 \times 10^{7}\ \mathrm{erg\,K^{-1}\,g^{-1}}$
$k =$	Boltzmann's constant	$= m_u \Re = 1.38 \times 10^{-16}\ \mathrm{erg\,K^{-1}} = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$
$a =$	radiation density constant	$= \dfrac{4\sigma}{c} = 7.56 \times 10^{-16}\ \mathrm{J\,m^{-3}\,K^{-4}}$
$\sigma =$	Stefan-Boltzmann constant	$= 5.67 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$
$c =$	speed of light	$= 2.998 \times 10^{10}\ \mathrm{cm\,s^{-1}}$
$h =$	Planck's constant	$= 6.62 \times 10^{-34}\ \mathrm{J\,s} = 2\pi\hbar$
$m_u =$	atomic mass unit (amu)	$= 1.66054 \times 10^{-24}\ \mathrm{g}\ \equiv 931.5\ \mathrm{MeV}$
$R_\odot =$	solar radius	$= 6.96 \times 10^{10}\ \mathrm{cm}$
$M_\odot =$	solar mass	$= 1.989 \times 10^{33}\ \mathrm{g}$
$L_\odot =$	solar luminosity	$= 3.86 \times 10^{33}\ \mathrm{erg\,s^{-1}}$
	1 eV	$= 1.602 \times 10^{-19}\ \mathrm{J}$
	1 J	$= 10^{7}\ \mathrm{erg}$
	1 Jy	$= 10^{-26}\ \mathrm{W\ m^{-2}\ Hz^{-1}}$
	1 AU	$= 1.50 \times 10^{11}\ \mathrm{m}$

Physical Formulae:

$E_\gamma = h\nu = hc/\lambda$

gravitational potential energy: $U = -\dfrac{GMm}{r}$

blackbody (Planck) function: $B_\nu(T) = \dfrac{2h\nu^3/c^2}{e^{h\nu/kT} - 1}$

redshift: $z = \frac{\Delta\lambda}{\lambda} = \frac{\lambda_{obs}}{\lambda_{emitted}} - 1$

Doppler shift: $\frac{v}{c} = z = \frac{\Delta\lambda}{\lambda_0}$ if $v \ll c$, otherwise $\frac{v}{c} = \frac{(1+z)^2-1}{1+(1+z)^2}$

ideal gas equation of state: $P = \frac{\rho k T}{\mu m_H}$

gravitational time dilation: $\Delta t_\infty = \Delta t_0 \left(1 - \frac{2GM}{Rc^2}\right)^{-1/2}$ & redshift $\nu_\infty = \nu_0 \left(1 - \frac{2GM}{Rc^2}\right)^{1/2}$

Schwarzschild Radius: $r_S = \frac{2GM}{c^2}$

Astronomical Formulae:

Flux of an object of luminosity L (radiating photons isotropically) at distance d: $F = \frac{L}{4\pi d^2}$

Stefan-Boltzmann law for luminosity of an object with radius R and temperature T : $L = 4\pi R^2 \sigma T^4$

Stefan-Boltzmann law for surface flux of an object with temperature T : $F = \sigma T^4$

Wien's law (with T in units of K): $\lambda_{\max} = 0.002897755\, T^{-1}$ m or alternatively, $\nu_{\max} = 5.88 \times 10^{10}\, T$ Hz

Rayleigh-Jeans approximation: $B_\nu = \frac{2\nu^2 kT}{c^2} = \frac{2kT}{\lambda^2}$

$m_1 - m_2 = -2.5 \log_{10}\left(\frac{F_1}{F_2}\right) = -2.5 \log_{10}\left(\frac{\mathcal{B}_1}{\mathcal{B}_2}\right)$

$m - M = 5 \log_{10}\left(\frac{d}{10\ \mathrm{pc}}\right)$

circular speed of a particle orbiting a mass M at a distance r: $V = \sqrt{\frac{GM}{r}}$

velocity of the Earth around the Sun: $V_E = 30\,\mathrm{km\,s^{-1}}$

Binary stars: $P^2 = \frac{4\pi^2 a^3}{G(M_1+M_2)}$

Statistics Formulae:

Gaussian distribution: $f(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} \exp\left[\dfrac{-(x-\mu)^2}{2\sigma^2}\right]$

Poisson distribution: $P_x = \frac{m^x \exp(-m)}{x!}$

Signal-to-noise ratio: $SNR = \frac{S}{\sqrt{S+B(1+A_S/A_B)}}$

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